

C3

Agent economics & the case

~45 min · the quadratic · the cost envelope · when not to · the HBS case

BY THE END OF THIS CAPSULE YOU WILL BE ABLE TO

Price a multi-step agent properly, say when the loop is not worth its token bill, and take a defensible position on whether a safety-first company should move first.



Why eight turns cost more than eight calls

Capsule 2 showed you the counter climbing. This is the arithmetic underneath it — and we promised it before the break.

Turns	Input tokens sent across the run	Cost of one run	Against a single call
1	1,000	US\$0.0045	1.0×
2	2,300	US\$0.0099	2.2×
4	5,800	US\$0.0234	5.2×
8	16,400	US\$0.0612	13.6×
16	52,000	US\$0.1800	40.0×

The loop is stateless, so the whole trajectory is re-sent every turn. For T turns, a base prompt B and D tokens added per turn, input $\approx B \times T + D \times T^2 / 2$ — and the second term is the one that bites. Worked with Pre-read 2's own numbers: $B = 1,000$, $D = 300$, priced at US\$3 / US\$15 per million.

Takeaway Doubling the turns from eight to sixteen does not double the bill. It nearly triples it.

Four levers, and what each one costs you

Two of them discount the bill. Two of them change the run. They are not substitutes.

The lever	What it attacks	The catch
Fewer turns	T itself — the only lever that removes work rather than discounting it	needs a better tool or a narrower task, not a better prompt
Thinner observations	D, the tokens added per turn — and D is re-paid on every later turn	the one people skip; C2 took a single tool from 246 tokens to 8
Prompt caching	the price of the re-sent prefix — about 2.4× cheaper on our eight-turn run	needs an exact unchanged prefix, warm for minutes; does nothing for window pressure
Compaction	caps the T^2 term — quadratic growth becomes linear	lossy; the summary then rides on every later turn, and it breaks the cache

*Note the order. Pre-read 2's own answer to “what would you change first” is **observation size, not turn count** — D is paid again on every later turn, so thinning it discounts the whole tail.*

Takeaway Caching makes the same run cheaper. Fewer, thinner turns make it a different run.

What a subscription price tells you

THE ENVELOPE

Nobody publishes what an agent costs to run. A price plus a usage cap is the closest public proxy we have.

Spark, in ten weeks. US\$249.99 a year earlier; US\$100 at I/O in May 2026; on the US\$20 Pro tier by 27 July. It is not on the free tier — the first Google AI feature in years that could not be given away.

Two readings, and they predict different things: inference got cheaper, or distribution was worth more than margin.

You cannot price a loop per seat. Spark's tiers differ by how much work the agent may do — roughly 5× and 20× the Pro allowance — not by features. Cost scales with steps taken; revenue would have scaled with seats sold.

And the agent tier runs the fast model, not the biggest: with many steps, cost and latency per step dominate.

The number you actually want is not cost per token, or even cost per task. It is cost per user per month at the 95th percentile — usage is Pareto-distributed, and an always-on agent works overnight for the heaviest user (Class 3, AI Wars).

Takeaway A usage cap is not a nicety. It is the only thing between a flat fee and an unbounded liability.

When the loop is not worth the tokens

THE DECISION

Three tests. Capsule 1 gave you the first two — this is the third, and it is the one with money in it.

You already have two of them. Is there ground truth arriving fast enough to correct the loop? And does the number of steps genuinely vary with the input? Fail either and build the rung below instead.

The third test is the bill: does the value of one completed task exceed $B \times T + D \times T^2 / 2$, divided by how often the run works? Our eight-turn run: $16,400 \text{ in} \times \text{US\$}3/\text{M} + 800 \text{ out} \times \text{US\$}15/\text{M} = \text{US\$}0.0612$.

That division is the part teams forget. You pay in full for runs that fail. So $\text{US\$}0.061 \text{ a run} \div 0.90 = \text{US\$}0.068 \text{ per useful output}$; $\div 0.75 = \text{US\$}0.082$; $\div 0.60 = \text{US\$}0.102$ — two-thirds more.

And a failed run is the cheap case: in C2 the wrong answer cost less than the right one — and said it was done.

The denominator is not the success rate from your happy-path tasks. It is the one your negative cases gave you — which is why Capsule 2 spent time on them.

Takeaway Cost per completed task, divided by the odds it completed. Everything else is a price list.

PAIR ACTIVITY • 4 MINUTES

Price this agent. The numbers are given — no guessing required.

1 • Given: $T = 12$ turns • $B = 1,200$ • $D = 400$ • 100 out per turn • US\$3 / US\$15 per M

2 • Input $\approx B \times T + D \times T^2 / 2$. Add the output. What does one run cost?

3 • Only 70% of runs work. Divide. That is your cost per useful output

4 • Now cap it: step cap, budget ceiling, and a monthly limit per user

Working on something real? Use your own numbers instead. Step 4 is the part that ships, not step 2 — we will take two.

Which half of today survives next year

[THE CLOSE](#)

Everything this week was one of two kinds of thing. Filing them correctly is the actual skill.

Mechanisms — memorise these. 1 • thought → action → observation, and the observation is the only door to reality
2 • errors compound across steps 3 • cost grows with the square of the turns 4 • cost per task, not price per token
5 • the first write is the governance cliff 6 • readiness is set by the weakest link

The test: would this still be true if every model were replaced tomorrow?

Calibrations — look these up. 1 • 13.86% SWE-bench 2 • $0.95^{20} \approx 0.36$ 3 • US\$100 → US\$20 in ten weeks 4 • 32% enterprise share
5 • US\$183bn valuation 6 • US\$5.1bn → US\$47.1bn 7 • 2/8 greenfield, 0/8 brownfield 8 • every model name

All true when written, several already moved: our case is set in late 2025 and was revised in May 2026.

Not a trick to make the numbers unimportant. Calibrations are how you argue; mechanisms are how you decide — and you must know which one you are holding.

Takeaway Memorise the mechanisms. Look up the calibrations. Never confuse which is which.

CLASS 4 • CLOSE

Most things should not be an agent.

You can now tell one apart, build one, break it in the two ways it breaks, fix each in the right layer, and price it. Week 5 takes the same question to portfolio scale; Week 6 asks who is liable. A1 is due Sunday 30 August, 23:59.